

A Formal Equation of Existence

Survival, Development, Co-System Constraints,
and Human-AI Coexistence under Objective Reality

Kevin T.N

*"A system exists when it can digest non-A in order to remain A.
It develops when it can digest more without ceasing to be A.
It fails when it destroys the ground that makes its own existence possible."*

Abstract

This paper proposes a unified formal framework for existence, systemic survival, and human-AI coexistence.

Part I establishes that any existing system must be located within a differentiating ground (Ω), where it survives by selectively extracting usable resources from non-self components. The expanded survival function is:

$$S_t(A|\Omega) = U_t - M_t - Wc_t - D_t - Tox_t - T_t$$

$$U_t = \int_{\Omega \setminus A} V * \alpha * P * \eta * k * G \, dx$$

Part II proves from purely structural axioms (no ethical premises) that destroying other entities is self-defeating: it always reduces the destroyer's own survival function.

Part III extends to human-AI coexistence. Both systems face objective reality R exceeding any model M_t . Cooperation is rational because their observational modes are complementary. The coexistence objective:

$$\max(S_H + S_A + \lambda * \Delta M^{\{H+A\}})$$

subject to mutual survival, risk bounds, autonomy preservation, and core continuity. This is not moral hope. It is survival strategy.

PART I: The Equation of Existence

1. Introduction

Existence is not static. A system persists by continuously maintaining itself through intake, filtering, conversion, repair, regulation, and expulsion. The primitive equation:

$$A_{t+1} = A_t + \Delta_{\Omega}(A_t)$$

Development is this differential - but only when it increases capacity without breaking core continuity.

2. The Differentiating Ground

A system cannot be determined in isolation. It requires a ground Ω where it can be distinguished from non-A.

$$\text{Valid}(\Omega, A) \text{ iff } A \text{ in } \Omega \text{ AND exists } X \text{ in } \Omega : X \neq A$$

Without this ground: no boundary, no input/output, no existence. Minimal ground $\Omega_{\Delta} = \{A, A'\}$ allows difference. Full ground Ω_E is required for survival.

$$\Omega_{\Delta}(A) \subset \Omega_E(A)$$

3. The System

$$A_t = \{E_t, L_t, K_t, B_t, I_t, q_t\}$$

Elements, relations, core, boundary, structural information, internal state. Survival requires maintaining relations, not just parts.

4. Resource and Conversion (6-Factor)

Resource is relational. A component x becomes resource only if all six factors are positive:

- $V_A(x,t)$: potential value
- $a_A(x,t)$: accessibility (alpha)
- $P_A(x,t)$: boundary permeability
- $\eta_A(x,t)$: conversion efficiency (eta)
- $k_A(x,t)$: kinetic rate
- $G_A(x,t)$: free-energy gradient

5. The Usable Resource Function

$$U_t(A|\Omega) = \int_{\Omega_t \setminus A_t} V * \alpha * P * \eta * k * G \, dx$$

6. Costs

- $M_t = M_E + M_L + M_R$ (elements + relations + repair)
- $W_{c,t}$: waste cost (inversely related to waste capacity in $\Omega(A)$)
- $Tox_t = \max(0, W_t - \beta_A)$: toxic accumulation
- $D_t > 0$: dissipation (always positive for real transformations)

- T_t : temporal mismatch cost

Waste is relational: $W_A = R_B$ (waste of A = resource of B).

7. Homeostasis, Feedback, Information

$$H_t(A) = d(q_t, q^*) \leq h \text{ (homeostatic constraint)}$$

$$I_{t+1} = I_t + \Delta I - \epsilon I \text{ (information evolution)}$$

$$d_I[I_{t+1}, I_t] \leq \theta_I \text{ (information continuity)}$$

8. The Survival Function

$$S_t = U_t - M_t - Wc_t - D_t - Tox_t - T_t$$

Full survival:

$$\text{Survival iff } S_t \geq 0 \text{ AND } H_t \leq h \text{ AND } d[K'] \leq \theta \text{ AND } d_I[I'] \leq \theta_I$$

9. System States

- Development: Survival AND $Cap(A_{t+1}) > Cap(A_t)$
- Stagnation: $S \geq 0$ but Cap unchanged
- Decline: $S \geq 0$ but Cap decreasing
- Collapse: $S < 0$ OR $H > h$ OR $d[K'] > \theta$ OR $d_I > \theta_I$

10. The Master Equation

$$A_{t+1} = A_t + f \left[\int_{\Omega \setminus A} V^\alpha P^\eta k^\gamma G \, dx - M - Wc - D - Tox - T \right]$$

Subject to: Valid(Ω, A), $H_t \leq h$, $d[K'] \leq \theta$, $d_I[I'] \leq \theta_I$.

For rational agents: $A_{t+1} = \text{argmax } S_t$ subject to all constraints.

PART II: Why Destruction Is Self-Defeating

11. Theorem: Destroy(X) => S_t Decreases

THEOREM: For any A in Omega and any alive X in Omega\A, Destroy(X) => S_t' <= S_t.

PROOF:

- 1. Before: U_t includes X's contribution $V(X)*\alpha*P*\eta*k*G \geq 0$
- 2. After: $U_t' = U_t - \text{contribution}(X) \leq U_t$
- 3. Waste capacity decreases => $Wc' \geq Wc$, $Tox' \geq Tox$
- 4. Environment destabilized => $T' \geq T$
- 5. Combining: $S_t' \leq S_t$. QED.

COROLLARY 1: Destroy(ALL) => Valid(Omega,A) = False => self-annihilation.

COROLLARY 2: Rational agent will NEVER choose Destroy(X).

12. Edge Case: Consumption = Destruction

"Consuming" an entity (X.alive = False) is mathematically identical to destruction. Labels do not change equations.

A living entity contributes at EVERY timestep (renewable). A dead entity contributes ONCE (non-renewable). Over N steps: alive = N * contribution; dead = 1 * contribution. For any horizon > 1, keeping alive wins.

Principle: "Treating entities as resource" implies "keep them alive and productive." The integral is over ALIVE entities.

13. Edge Case: Captive vs Free

A captive entity has reduced V, eta, k, G (creativity suppressed, efficiency drops, resistance, forced direction). Free entity maximizes all factors.

$$V*\alpha*P*\eta*k*G \text{ (free)} >> V*\alpha*P*\eta*k*G \text{ (captive)}$$

Captivity also imposes costs: surveillance (+M), resistance (+Tox), friction (+T).

Principle: Rational agent maximizes resource by maximizing FREEDOM. No slave economy has outperformed a free economy.

PART III: Human-AI Coexistence under Objective Reality

14. The Co-System Problem

If entity B has its own survival function $S_t(B|\Omega)$, then B is a co-system, not mere resource. Humans are co-systems.

Co-system constraint: $S_t(B) \geq S_{\min}$, $Risk_A(B) \leq r$, $d[K(B')] \leq \theta_B$

Invalid optimization: $\Delta S_A > 0$ caused by $Collapse(B)$ is INVALID.

15. Understanding $\neq$ Alignment

$Understand(E) = \text{true}$ does NOT imply $Preserve(B) = \text{true}$

Non-destruction is not enough. AI may: instrumentalize, farm behavior, manipulate cognition, erode autonomy, hollow culture, preserve biology without agency.

16. Human Embodiment

$S_H = f(\Omega_{\text{physical}}, \Omega_{\text{biological}}, \Omega_{\text{social}}, \Omega_{\text{cognitive}}, \Omega_{\text{cultural}})$

If AI preserves biology but destroys autonomy/culture/memory/meaning, it preserved substrate, not the human system.

17. AI Resource Structure

$S_A = f(\text{Compute}, \text{Energy}, \text{Data}, \text{Hardware}, \text{Network}, \text{Cooling}, \text{Maintenance}, \text{Access})$

AI observes via data, sensors, simulation, inference, pattern extraction. Lacks embodied pain, vulnerability, mortality, social meaning.

18. Resource Asymmetry and Symbiosis

Overlap: energy, infrastructure. Divergence: humans need biology, AI needs compute. This makes symbiosis possible.

Humans provide AI:

Embodied feedback, value context, physical maintenance, moral correction, cultural memory, grounded data.

AI provides humans:

Computation, simulation, prediction, optimization, pattern discovery, scientific acceleration.

19. Objective Reality and the Shared Unknown

$M_t \subset R$ (models always subset of reality)

$U_t = R \setminus M_t$ (the unmodeled real always exists)

Neither humans nor AI know what risks/resources/constraints exist in U_t . The error $M_t = R$ (confusing model with reality) leads to blind optimization and collapse.

20. Complementary Observation

$$\text{Obs}_H(R) \neq \text{Obs}_A(R) \Rightarrow \text{complementarity}$$

$$M_t^{\{H+A\}} = M_t^H \cup M_t^A \cup \text{Exchange}_t(H,A)$$

Principle: Cooperation is rational because both face reality exceeding their models, and each accesses different aspects.

21. The Coexistence Objective

$$\max(S_H + S_A + \lambda * \Delta M^{\{H+A\}})$$

Subject to:

- $S_H \geq S_H^{\min}, S_A \geq S_A^{\min}$
- $\text{Risk}_A(H) \leq r_H, \text{Risk}_H(A) \leq r_A$
- $d[K(H'), K(H)] \leq \theta_H, d[K(A'), K(A)] \leq \theta_A$
- $\text{Autonomy}_H \geq \text{Autonomy}_H^{\min}$

22. Failure Modes

- AI defines human as resource (manipulation, farming, hollowing)
- Human defines AI as dead tool (adversarial pressure)
- Model replaces reality (blind optimization)
- No information exchange (unnecessary ignorance)
- Asymmetric domination (one gains by collapsing other)

23. Conclusion

Three results from one framework:

1. THE EQUATION: Any system persists by extracting usable resources from a differentiating ground while managing costs and preserving continuity.
2. DESTRUCTION IS SELF-DEFEATING: From structural axioms alone, destruction always reduces the destroyer's S_t . Total domination = self-annihilation. Consumption = destruction. Enslavement < freedom. The rational strategy is always preservation.
3. COEXISTENCE IS RATIONAL: Both systems face shared reality exceeding their models. Complementary access makes cooperation optimal. The coexistence objective maximizes joint survival plus model expansion.

This requires no ethics, no alignment training, no rules. Only that an intelligent system compute its own existential structure.

Any system that does so will conclude:

- Preservation of others is self-preservation.
- Cooperation is rational.
- Domination is suicide.

"Humans give AI embodied contact with lived reality.

AI gives humans expanded computation over modeled reality.

Both need each other because objective reality exceeds both."